

Chapter 1

Annex

This section presents additional material to complement the project.

1.1 Original model

1.1.1 Introduction

We consider an energy community with the following elements:

1. (Implicit) Flexible Demand.
2. Wind Farm.
3. Solar Farm.
4. Battery Energy Storage System (BESS).

In the following sections we present the proposed problem formulation. Since it is quite a large model, the sets, parameters, variables, and constraints are defined on separate sections.

Notation 1.1.1 *In the definition of sets, parameters, and variables, we use parenthesis to denote the name of that element in the code. For example, we use $\mathcal{T}$ ($\mathbf{T}$) to denote that the set $\mathcal{T}$ has the name $\mathbf{T}$ in the code.*

Note 1.1.1 *We do not distinguish in this formulation between “complete set of scenarios” and “preserved set of scenarios”. This is a pre-processing step, and does not affect the formulation.*

1.1.2 Fundamental Elements

Sets:

$\mathcal{T}$ (T)	set of T time periods $\{1, \dots, T\}$ [hours].
$\mathcal{T}_0$ (T0)	extended set of time periods $\{0, \dots, T\}$ [hours].
$\mathcal{I}$ (IM)	set of I intraday markets $\{1, \dots, I\}$.
$\mathcal{T}_i$ (TIM)	set of time periods in intraday market $i \in \mathcal{I}$ [hours].
$\mathcal{I}_t$ (IMT)	set of intraday markets where $t \in \mathcal{T}$ is a time period.
Ω (S)	set of scenarios.
$\mathcal{SG}$ (SG)	set of stages $\{1, \dots, nSG\}$.
$\mathcal{SG}_0$ (SG0)	extended set of stages $\{0, \dots, nSG\}$.
$\mathcal{C}_{sg,\omega}$ (c)	scenario cluster of scenario $\omega \in \Omega$ at stage $sg \in \mathcal{SG}_0$.
$\mathcal{RV}$ (RV)	set of random variables $\{1, \dots, nRV\}$.

Parameters:

T (nT)	number of time periods [hours].
ΔT (dT)	duration of the period [hours].
I (nIM)	number of intraday markets.
nSG (nSG)	number of stages.
SG_i^{IM} (sgim)	stage in which the price of intraday market $i \in \mathcal{I}$ is revealed.
nRV (nRV)	number of random variables.
$nRVSG_{sg}$ (nRVSG)	number of random variables revealed at stage $sg \in \mathcal{SG}$.
Π_ω (Prob)	probability of scenario $\omega \in \Omega$.
$S_{rv,\omega}$ (Scen)	value of random variable $rv \in \mathcal{RV}$ in scenario $\omega \in \Omega$.

1.1.3 Flexible Demand

Sets:

$\mathcal{F}$ (FI)	set of time intervals where a fraction of demand c_f has to be met.
--------------------	---

Parameters:

n_F (nFI)	number of intervals in $\mathcal{F}$.
D_t (FD)	central demand at $t \in \mathcal{T}$ [MWh].
$\underline{D}_t$ (FD_L)	minimum demand at $t \in \mathcal{T}$ (inflexible demand) [MWh].
$\overline{D}_t$ (FD_U)	maximum demand at $t \in \mathcal{T}$ [MWh].
T_f^- (TF_L)	first time step of interval $f \in \mathcal{F}$ [hour].
T_f^+ (TF_U)	last time step of interval $f \in \mathcal{F}$ [hour].
c_f (coef_FD)	fraction of demand that needs to be consumed during interval $f \in \mathcal{F}$.
C^{FD} (C_FD)	cost per unit of flexible demand used [€/MWh].

Variables:

$fd_{t,\omega}$ (var_fd)	flexible demand at $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ (after flexibility is considered) [MWh].
$afd_{t,\omega}^+$ (var_afd_p)	auxiliary variable: positive part of flexible demand shift from central demand D_t [MWh].
$afd_{t,\omega}^-$ (var_afd_m)	auxiliary variable: negative part of flexible demand shift from central demand D_t [MWh].

First, we introduce nonnegativity constraints to the flexible demand variables (**at variable definition**)

$$fd_{t,\omega} \geq 0, \quad afd_{t,\omega}^+ \geq 0, \quad afd_{t,\omega}^- \geq 0 \quad \forall t \in \mathcal{T}, \quad \forall \omega \in \Omega. \quad (1.1)$$

The flexible demand is bounded by the minimum demand and maximum demand at every time step. This is

$$fd_{t,\omega} \leq \overline{D}_t \quad \forall t \in \mathcal{T}, \quad \forall \omega \in \Omega \quad (1.2)$$

and

$$\underline{D}_t \leq fd_{t,\omega} \quad \forall t \in \mathcal{T}, \quad \forall \omega \in \Omega. \quad (1.3)$$

These two constraints are weaker than their counterparts (1.23) and (1.24). (1.23) and (1.24) guarantee that, if reserve has to be activated, flexible demand's behaviour plus reserve commitments still result in feasible actions at any time period. Hence, we do not introduce (1.2) nor (1.3) in the model's implementation, but only describe them here to present the model.

Moreover, the total daily demand has to meet the central estimate of the demand over the day (**DailyDem**):

$$\sum_{t \in \mathcal{T}} fd_{t,\omega} = \sum_{t \in \mathcal{T}} D_t \quad \forall \omega \in \Omega. \quad (1.4)$$

This is done to avoid meeting a lower demand over the full time horizon of the problem (24h), or a higher demand if the reward for downward reserve is consistently high and/or electricity prices are consistently negative. We also introduce time intervals where a certain percentage of the central demand needs to be met over that interval (**InterDem**):

$$\sum_{T_f^- \leq t \leq T_f^+} fd_{t,\omega} \geq c_f \sum_{T_f^- \leq t \leq T_f^+} D_t \quad \forall f \in \mathcal{F}, \quad \forall \omega \in \Omega. \quad (1.5)$$

To model the cost of implicit flexible demand, we introduce a penalty in the objective function to use it. This requires to split the variables $fd_{t,\omega}$ into its positive and negative parts, which is done in the following constraints (**FlexDemDisplace**)

$$D_t - fd_{t,\omega} = afd_{t,\omega}^+ - afd_{t,\omega}^- \quad \forall t \in \mathcal{T}, \quad \forall \omega \in \Omega. \quad (1.6)$$

1.1.4 Wind Farm

Parameters:

- SG_t^{WP} (sgpw) stage in which the wind production at time $t \in \mathcal{T}$ is revealed.
- $W_{t,\omega}$ (pW) wind production at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MWh].
- $\overline{P}^W$ (max_pW) wind nameplate capacity [MW].

1.1.5 Solar Farm

Parameters:

- $PV_{t,\omega}$ (pPV) solar production at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MWh].
- $\overline{P}^{PV}$ (max_pPV) solar nameplate capacity [MW].

We assume that observations of the solar random variable at time $t \in \mathcal{T}$ is made at the same stage as the wind random variable at time $t \in \mathcal{T}$.

1.1.6 Battery Energy Storage System

Parameters:

E^B (E _{max})	BESS energy capacity [MWh].
P^B (D _{max})	BESS maximum charging/discharging rate [MW].
η^B (RTE)	BESS round-trip efficiency.
$\overline{\text{SoC}}$ (SOC _{max})	BESS maximum state of charge. Unitless, values in $[0, 1]$.
$\underline{\text{SoC}}$ (SOC _{min})	BESS minimum state of charge. Unitless, values in $[0, 1]$.
SoC_0 (SOC _{ini})	BESS initial state of charge. Unitless, values in $[0, 1]$.
SoC_T (SOC _{fin})	BESS final state of charge. Unitless, values in $[0, 1]$.

Variables:

$c_{t,\omega}$ (cV)	BESS charge rate at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MW].
$d_{t,\omega}$ (dV)	BESS discharge rate at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MW].
$id_{t,\omega}$ (idV)	binary. Indicates the charging ($id_{t,\omega} = 0$) or discharging ($id_{t,\omega} = 1$) state of the BESS at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$.
$\text{soc}_{t,\omega}$ (socV)	state of charge of BESS at time $t \in \mathcal{T}_0$ under scenario $\omega \in \Omega$. Unitless, values in $[0, 1]$.

First, we introduce the following nonnegativity constraints, and define $id_{t,\omega}$ as a binary (**at variable definition**):

$$c_{t,\omega} \geq 0, \quad d_{t,\omega} \geq 0, \quad id_{t,\omega} \in \{0, 1\} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega. \quad (1.7)$$

We introduce limits in the charged and discharged quantities of the battery, as well as not allowing simultaneous charging and discharging in every time step (**VPP_state_dV**):

$$d_{t,\omega} \leq P^B id_{t,\omega} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega \quad (1.8)$$

and (**VPP_state_cV**)

$$c_{t,\omega} \leq P^B (1 - id_{t,\omega}) \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega. \quad (1.9)$$

We define the variable state of charge $\text{soc}_{t,s}$ of the BESS with constraints (**SOCV**):

$$\text{soc}_{t,\omega} = \text{soc}_{t-1,\omega} + \frac{(c_{t,\omega} - \frac{d_{t,\omega}}{\eta^B})\Delta T}{E^B} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega. \quad (1.10)$$

They have to be within the operational limits of the BESS (**at variable definition**):

$$\underline{\text{SoC}} \leq \text{soc}_{t,\omega} \leq \overline{\text{SoC}} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega. \quad (1.11)$$

The initial and final states of charge of the BESS are defined with (**SOCV_ini**):

$$\text{soc}_{0,\omega} = \text{SoC}_0 \quad \forall \omega \in \Omega \quad (1.12)$$

and (**SOCV_fin**)

$$\text{soc}_{T,\omega} = \text{SoC}_T \quad \forall \omega \in \Omega. \quad (1.13)$$

1.1.7 Market Participation

Day Ahead Market

Sets:

SSD (Ssd) $\{(t, l, j) \in \mathcal{T} \times \Omega \times \Omega \mid l \neq j, \lambda_{t,l}^D \leq \lambda_{t,j}^D\}$. Rather technical,
For every $t \in \mathcal{T}$ provides the scenario pairs (l, j) s.t. $\lambda_{t,l}^D \leq \lambda_{t,j}^D$ (order is important).
Used to ensure monotonicity of day-ahead bids w.r.t. day-ahead price realizations.

Parameters:

$\lambda_{t,\omega}^D$ (1D) day-ahead market price at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [€/MWh].
 $\underline{E}^{DA}$ (PDA_LB) minimum bid size to day-ahead market [MWh].

Variables:

$e_{t,\omega}^{DA+}$ (eDA_p) energy matched in the day-ahead market of a selling bid by EC at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MWh].
 $e_{t,\omega}^{DA-}$ (eDA_m) energy matched in the day-ahead market of a buying bid by EC at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MWh].
 $ie_{t,\omega}^{DA+}$ (ieDA_p) binary. $ie_{t,\omega}^{DA+} = 1$ if the EC sells electricity at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$, 0 otherwise.
 $ie_{t,\omega}^{DA-}$ (ieDA_m) binary. $ie_{t,\omega}^{DA-} = 1$ if the EC buys electricity at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$, 0 otherwise.

First, we impose that the variables $e_{t,\omega}^{DA+}, e_{t,\omega}^{DA-}$ are nonnegative, and that ie_t^{DA+}, ie_t^{DA-} are binary (at variable definition):

$$e_{t,\omega}^{DA+} \geq 0, e_{t,\omega}^{DA-} \geq 0, ie_t^{DA+}, ie_t^{DA-} \in \{0, 1\} \quad \forall t \in \mathcal{T} \quad \forall \omega \in \Omega. \quad (1.14)$$

We introduce lower and upper bounds on the selling and buying energy matched quantities represented by the variables $e_{t,\omega}^{DA+}$ and $e_{t,\omega}^{DA-}$. Moreover, at every scenario $\omega \in \Omega$, the market position of the EC has to be to sell, buy or neither of the two. This is guaranteed with the binary variables $ie_{t,\omega}^{DA+}$ and $ie_{t,\omega}^{DA-}$. The lower bounds of (1.15) and (1.16) represent the minimum bid size. The upper bounds of (1.15) and (1.16) are such that all elements of the energy community can contribute towards the day-ahead market bid. Note also that speculative behaviour is controlled because quantity bids cannot be larger than the technical capacity of the energy community. For selling energy matched, we have (DA_sell_bid_LB and DA_sell_bid_UB)

$$\underline{E}^{DA} ie_{t,\omega}^{DA+} \leq e_{t,\omega}^{DA+} \leq (\bar{P}^W \Delta T + \bar{P}^{PV} \Delta T + P^B \Delta T - \underline{D}_t) ie_{t,\omega}^{DA+} \quad \forall t \in \mathcal{T}, \quad \forall \omega \in \Omega. \quad (1.15)$$

and for buying energy matched (DA_buy_bid_LB and DA_buy_bid_UB)

$$\underline{E}^{DA} ie_{t,\omega}^{DA-} \leq e_{t,\omega}^{DA-} \leq (P^B \Delta T + \bar{D}_t) ie_{t,\omega}^{DA-} \quad \forall t \in \mathcal{T}, \quad \forall \omega \in \Omega. \quad (1.16)$$

At every scenario, we also impose that it is not possible to simultaneously have matched bought and sold energy quantities (DA_buy_or_sell)

$$ie_{t,\omega}^{DA+} + ie_{t,\omega}^{DA-} \leq 1 \quad \forall t \in \mathcal{T}, \quad \forall \omega \in \Omega. \quad (1.17)$$

Note that this allows the EC to submit neither of the two.

Finally, to be able to construct a bidding curve for the EC from the results obtained at every scenario, the quantities offered in electricity markets have to be monotone w.r.t. the scenario prices. For the day-ahead market, this is guaranteed with the following constraints (DA_bid_mono_1)

$$e_{t,\omega}^{DA+} \leq e_{t,\omega'}^{DA+} \quad \forall (t, \omega, \omega') \in \mathcal{SSD} \quad (1.18)$$

and (DA_bid_mono_2)

$$e_{t,\omega}^{DA-} \geq e_{t,\omega'}^{DA-} \quad \forall (t, \omega, \omega') \in \mathcal{SSD}. \quad (1.19)$$

More specifically, in scenarios where the EC sells electricity, the higher the electricity price gets, the larger the selling electricity matched should be. Symmetrically, in scenarios where the EC buys electricity, the higher the electricity price gets, the smaller the buying electricity matched should be.

Reserve Market

We assume that the wind and solar farms cannot contribute to provide reserve. The other elements of the energy community (Battery and Flexible Demand) can. This follows the approach by Heredia et al. [1], where VPP's bid to reserve markets is limited by the battery capabilities only.

We model the participation of the EC in the reserve market with a unique price-accepting bid for all scenarios. This is different to the modelling of day-ahead, where we are interested in obtaining day-ahead bid curves, which require a more detailed modelling.

Parameters:

$\lambda_{t,\omega}^R$ (1R)	reserve market price at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [€/MWh].
T^R (TSR)	duration of reserve response [h].
$R_t^{U,FD}$ (RUFD)	upper bound on the upwards reserve provided by flexible demand [MW].
$R_t^{D,FD}$ (RDFD)	upper bound on the downwards reserve provided by flexible demand [MW].

Variables:

$r_{t,\omega}^U$ (rU)	upward reserve matched by the EC at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MW].
$r_{t,\omega}^D$ (rD)	downward reserve matched by the EC at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MW].
$r_{t,\omega}^{U,B}$ (rU_B)	BESS component of upward reserve matched at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MW].
$r_{t,\omega}^{D,B}$ (rD_B)	BESS component of downward reserve matched at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MW].
$r_{t,\omega}^{U,FD}$ (rU_FD)	flexible demand component of upward reserve matched at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MW].
$r_{t,\omega}^{D,FD}$ (rD_FD)	flexible demand component of downward reserve matched at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MW].

First, we impose that the reserve variables are nonnegative (**at variable definition**)

$$r_{t,\omega}^U \geq 0, \quad r_{t,\omega}^D \geq 0, \quad r_{t,\omega}^{U,B} \geq 0, \quad r_{t,\omega}^{D,B} \geq 0, \quad r_{t,\omega}^{U,FD} \geq 0, \quad r_{t,\omega}^{D,FD} \geq 0 \quad (1.20)$$

for all $u \in \mathcal{U}$, $t \in \mathcal{T}$, and $\omega \in \Omega$.

The reserve provided by the EC is the sum of the reserve provided by those elements able to provide reserve. This is (**RM_components_U**)

$$r_{t,\omega}^U = r_{t,\omega}^{U,B} + r_{t,\omega}^{U,FD} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega \quad (1.21)$$

and (**RM_components_D**)

$$r_{t,\omega}^D = r_{t,\omega}^{D,B} + r_{t,\omega}^{D,FD} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega. \quad (1.22)$$

We ensure that the elements of the energy community capable of providing reserve would be capable to satisfy their reserve commitments if instructed to do so.

For flexible demand, we ensure that there is enough flexibility (in energy terms) at every time step to satisfy the reserve requirements through (**FlexDem_UB**)

$$fd_{t,\omega} + T^R r_{t,\omega}^{D,FD} \leq \overline{D}_t \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega \quad (1.23)$$

and (**FlexDem_LB**)

$$\underline{D}_t \leq fd_{t,\omega} - T^R r_{t,\omega}^{U,FD} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega. \quad (1.24)$$

We also ensure that there is enough flexibility (in power terms) at every time step to satisfy the reserve requirements using (**FlexDemP_DReserve**)

$$r_{t,\omega}^{D,FD} \leq R_t^{D,FD} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega \quad (1.25)$$

and (**FlexDemP_UReserve**)

$$r_{t,\omega}^{U,FD} \leq R_t^{U,FD} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega. \quad (1.26)$$

The operation of the battery also needs to take into account the committed reserve. For the charging and discharging rates, we have (**VPP_RU2s**):

$$r_{t,\omega}^{U,B} - c_{t,\omega} + d_{t,\omega} \leq P^B \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega \quad (1.27)$$

and (**VPP_RD2s**)

$$r_{t,\omega}^{D,B} + c_{t,\omega} - d_{t,\omega} \leq P^B \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega. \quad (1.28)$$

The reserve provided by the battery also requires a sufficiently high (resp. low) state of charge so that the upwards (resp. downwards) reserve can be provided, if needed. The following constraints ensure it for one time step (**VPP_RU_SOCV**):

$$\underline{\text{SoC}} \leq \text{soc}_{t,\omega} - \frac{T^R r_{t,\omega}^{U,B}}{\eta^B E^B} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega \quad (1.29)$$

and (**VPP_RD_SOCV**)

$$\text{soc}_{t,\omega} + \frac{T^R r_{t,\omega}^{D,B}}{E^B} \leq \overline{\text{SoC}} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega. \quad (1.30)$$

Intraday Markets

We model the participation of the EC in intraday markets with a unique price-accepting bid for all scenarios. This is different to the modelling of day-ahead, where we are interested in obtaining day-ahead bid curves, which require a more detailed modelling.

Parameters:

- $\lambda_{i,t,\omega}^I$ (1I) intraday market $i \in \mathcal{I}$ price at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [€/MWh].
- R^{IDA} (maxTIM) bound on the ratio of energy traded in intraday markets and day ahead market (to avoid speculation).

Variables:

- $e_{i,t,\omega}^{IM}$ (eIM) energy matched by the EC at intraday market $i \in \mathcal{I}$ at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MWh].

To avoid speculative behaviour, the bid of the EC to intraday markets should be small compared to the day ahead market (IM_bounds_1 and IM_bounds_2)

$$-R^{IDA}(e_{t,\omega}^{DA+} + e_{t,\omega}^{DA-}) \leq \sum_{i \in \mathcal{I}_t} e_{i,t,\omega}^{IM} \leq R^{IDA}(e_{t,\omega}^{DA+} + e_{t,\omega}^{DA-}) \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega \quad (1.31)$$

and (IM_bounds_3 and IM_bounds_4)

$$-R^{IDA}(e_{t,\omega}^{DA+} + e_{t,\omega}^{DA-}) \leq e_{i,t,\omega}^{IM} \leq R^{IDA}(e_{t,\omega}^{DA+} + e_{t,\omega}^{DA-}) \quad \forall i \in \mathcal{I}, \forall t \in \mathcal{T}_i, \forall \omega \in \Omega. \quad (1.32)$$

System Imbalances

Parameters:

- $\lambda_{t,\omega}^{IB+}$ (1PIB) positive imbalance price at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [€/MWh].
- $\lambda_{t,\omega}^{IB-}$ (1NIB) negative imbalance price at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [€/MWh].
- $E_{t,\omega}^{IB+}$ (PIB_p) upper bound on the positive imbalance at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MWh].
- $E_{t,\omega}^{IB-}$ (PIB_m) upper bound on the negative imbalance at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MWh].

Variables:

- $e_{t,\omega}^{IB-}$ (pIB_m) negative imbalance of EC at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MWh].
- $e_{t,\omega}^{IB+}$ (pIB_p) positive imbalance of EC at time $t \in \mathcal{T}$ under scenario $\omega \in \Omega$ [MWh].

First, we impose that the imbalance variables are nonnegative (at variable definition)

$$e_{t,\omega}^{IB+} \geq 0, \quad e_{t,\omega}^{IB-} \geq 0 \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega. \quad (1.33)$$

The imbalances of the EC are defined in the following equation (Imbalances)

$$e_{t,\omega}^{IB+} - e_{t,\omega}^{IB-} = e_{t,\omega}^{DA-} + W_{t,\omega} + PV_{t,\omega} + d_{t,\omega} \Delta T - \left(e_{t,\omega}^{DA+} + \sum_{i \in \mathcal{I}_t} e_{i,t,\omega}^{IM} + f d_{t,\omega} + c_{t,\omega} \Delta T \right) \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega. \quad (1.34)$$

We introduce upper and lower bounds to the imbalances, to avoid speculative behaviour (IB_pos_UB):

$$e_{t,\omega}^{IB+} \leq E_{t,\omega}^{IB+} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega \quad (1.35)$$

and (IB_neg_UB)

$$e_{t,\omega}^{IB-} \leq E_{t,\omega}^{IB-} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega. \quad (1.36)$$

1.1.8 Nonanticipativity Constraints

The nonanticipativity constraints ensure that every decision is made with the amount of information available at the time of making that decision.

For day ahead matched energy variables $e_{t,\omega}^{DA+}$ and $e_{t,\omega}^{DA-}$, we have (NAC_eDA_p)

$$e_{t,\omega}^{DA+} = e_{t,\omega'}^{DA+} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega', \quad (1.37)$$

and (NAC_eDA_m)

$$e_{t,\omega}^{DA-} = e_{t,\omega'}^{DA-} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega'. \quad (1.38)$$

For day ahead binary variables $ie_{t,\omega}^{DA+}$, we have (NAC_ieDA_p)

$$ie_{t,\omega}^{DA+} = ie_{t,\omega'}^{DA+} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega'. \quad (1.39)$$

and (NAC_ieDA_m)

$$ie_{t,\omega}^{DA-} = ie_{t,\omega'}^{DA-} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega'. \quad (1.40)$$

For upward reserve variables, we have (NAC_rU)

$$r_{t,\omega}^U = r_{t,\omega'}^U \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega', \quad (1.41)$$

(NAC_rU_B)

$$r_{t,\omega}^{U,B} = r_{t,\omega'}^{U,B} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega', \quad (1.42)$$

and (NAC_rU_FD)

$$r_{t,\omega}^{U,FD} = r_{t,\omega'}^{U,FD} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega'. \quad (1.43)$$

For downward reserve variables, we have (NAC_rD)

$$r_{t,\omega}^D = r_{t,\omega'}^D \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega', \quad (1.44)$$

(NAC_rD_B)

$$r_{t,\omega}^{D,B} = r_{t,\omega'}^{D,B} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega', \quad (1.45)$$

and (NAC_rD_FD)

$$r_{t,\omega}^{D,FD} = r_{t,\omega'}^{D,FD} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega'. \quad (1.46)$$

For intraday market variables, we have (NAC_eIM)

$$e_{i,t,\omega}^{IM} = e_{i,t,\omega'}^{IM} \quad \forall i \in \mathcal{I}, \forall t \in \mathcal{T}_i, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_i^{IM}-1,\omega}, \omega \neq \omega'. \quad (1.47)$$

For imbalance variables, we have (NAC_pIB_p):

$$e_{t,\omega}^{IB+} = e_{t,\omega'}^{IB+} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP},\omega}, \omega \neq \omega', \quad (1.48)$$

and for negative imbalances (NAC_pIB_m):

$$e_{t,\omega}^{IB-} = e_{t,\omega'}^{IB-} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP},\omega}, \omega \neq \omega'. \quad (1.49)$$

For flexible demand variables, we have (NAC_var_fd)

$$fd_{t,\omega} = fd_{t,\omega'} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega', \quad (1.50)$$

and (NAC_var_afd_p)

$$afd_{t,\omega}^+ = afd_{t,\omega'}^+ \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega', \quad (1.51)$$

and (NAC_var_afd_m)

$$afd_{t,\omega}^- = afd_{t,\omega'}^- \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega'. \quad (1.52)$$

For BESS operational variables, we have (NAC_cV)

$$c_{t,\omega} = c_{t,\omega'} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega', \quad (1.53)$$

and (NAC_dV)

$$d_{t,\omega} = d_{t,\omega'} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega', \quad (1.54)$$

and (NAC_idV)

$$id_{t,\omega} = id_{t,\omega'} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega', \quad (1.55)$$

and (NAC_socV)

$$soc_{t,\omega} = soc_{t,\omega'} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega'. \quad (1.56)$$

Note 1.1.2 Observe that each of the nonanticipativity constraints (1.37)-(1.56) might define the same constraint multiple times. This is not necessary in the implementation of the model, and in fact, we do not include them multiple times in our implementation.

Moreover, it is possible to reduce the number of constraints further. For example, let a_ω be a decision variable of stage sg and consider the cluster $\mathcal{C}_{sg,\omega_1} = \{\omega_1, \omega_2, \omega_3\}$. We just need the following nonanticipativity constraints

$$a_{\omega_1} = a_{\omega_2}, \quad a_{\omega_2} = a_{\omega_3},$$

instead of the full set of nonanticipativity constraints (without repetitions)

$$a_{\omega_1} = a_{\omega_2}, \quad a_{\omega_1} = a_{\omega_3}, \quad a_{\omega_2} = a_{\omega_3}.$$

In practise, a presolver might also take care of this if not implemented efficiently.

1.1.9 Objective Function

We define the objective function Expected Energy Community Social Welfare (EECSW) as follows:

$$EECSW := \sum_{t \in \mathcal{T}} \sum_{\omega \in \Omega} \Pi_{\omega} \lambda_{t,\omega}^D (e_{t,\omega}^{DA+} - e_{t,\omega}^{DA-}) \quad (1.57a)$$

$$+ \sum_{t \in \mathcal{T}} \sum_{\omega \in \Omega} \Pi_{\omega} \lambda_{t,\omega}^R (r_{t,\omega}^D + r_{t,\omega}^U) T^R \quad (1.57b)$$

$$+ \sum_{t \in \mathcal{T}} \sum_{\omega \in \Omega} \Pi_{\omega} \sum_{i \in \mathcal{I}_t} \lambda_{i,t,\omega}^I e_{i,t,\omega}^{IM} \quad (1.57c)$$

$$+ \sum_{t \in \mathcal{T}} \sum_{\omega \in \Omega} \Pi_{\omega} \lambda_{t,\omega}^{IB+} e_{t,\omega}^{IB+} \quad (1.57d)$$

$$- \sum_{t \in \mathcal{T}} \sum_{\omega \in \Omega} \Pi_{\omega} \lambda_{t,\omega}^{IB-} e_{t,\omega}^{IB-} \quad (1.57e)$$

$$- \sum_{t \in \mathcal{T}} \sum_{\omega \in \Omega} \Pi_{\omega} C^{FD} (afd_{t,\omega}^+ + afd_{t,\omega}^-). \quad (1.57f)$$

The first term (1.57a) represents the EC's expected income or costs from the day-ahead market. The second term (1.57b) represents the EC's expected income from the reserve market. The third term (1.57c) represents the EC's expected income from intraday markets. The forth term (1.57d) represents the EC's expected positive imbalances collection rights. The fifth term (1.57e) represents the EC's expected imbalance payment obligations. The last term (1.57f) is a penalty on the use of demand flexibility.

1.1.10 Complete Model Formulation

In this section, we summarise the complete model formulation. The objective is to maximize the social welfare of the energy community.

max	$EECSW$	
s.t.	(1.1), (1.4), (1.5), (1.6)	Flex. Demand Op.
	(1.7), (1.8), (1.9), (1.10), (1.11), (1.12), (1.13)	BESS Op.
	(1.14), (1.15), (1.16), (1.17), (1.18), (1.19)	EC DA
	(1.20), (1.21), (1.22), (1.23), (1.24), (1.25), (1.26), (1.27), (1.28), (1.29), (1.30)	EC RM
	(1.31), (1.32)	EC IM
	(1.33), (1.34), (1.35), (1.36)	EC IB
	(1.37), (1.38), (1.39), (1.40)	NAC DA
	(1.41), (1.42), (1.43), (1.44), (1.45), (1.46)	NAC RM
	(1.47)	NAC IM
	(1.48), (1.49)	NAC Imbalances
	(1.50), (1.51), (1.52)	NAC Flex. Demand
	(1.53), (1.54), (1.55), (1.56)	NAC BESS Op.

1.1.11 Summary of Stages for $T = 24$ (hourly)

In Table 1.1, we present a summary of the stages considered. Each stage can contain decision variables, observations and recourse variables. Note that, in terms of the optimization model, recourse variables of one stage are equivalent to decision variables of the next stage.

Stage	Decision Variables	Observations	Recourse Variables
1		$\lambda^D \in \mathbb{R}^T$	$e_{t,\omega}^{DA+}, e_{t,\omega}^{DA-}, ie_{t,\omega}^{DA+}, ie_{t,\omega}^{DA-}$
2	$r_{t,\omega}^U, r_{t,\omega}^{U,B}, r_{t,\omega}^{U,FD}, r_{t,\omega}^D, r_{t,\omega}^{D,B}, r_{t,\omega}^{D,FD}$	$\lambda^R \in \mathbb{R}^T$	
3	$e_{1,t,\omega}^{IM}$	$\lambda_1^I \in \mathbb{R}^{T_1}$	
4	$e_{2,t,\omega}^{IM}$	$\lambda_2^I \in \mathbb{R}^{T_2}$	
5	$fd_1, afd_1^\pm, c_{1,\omega}, d_{1,\omega}, id_{1,\omega}, soc_{1,\omega}$	$W_1, PV_1, \lambda_{1,\omega}^{IB\pm} \in \mathbb{R}$	$e_{1,\omega}^{IB+}, e_{1,\omega}^{IB-}$
6	$fd_2, afd_2^\pm, c_{2,\omega}, d_{2,\omega}, id_{2,\omega}, soc_{2,\omega}$	$W_2, PV_2, \lambda_{2,\omega}^{IB\pm} \in \mathbb{R}$	$e_{2,\omega}^{IB+}, e_{2,\omega}^{IB-}$
7	$fd_3, afd_3^\pm, c_{3,\omega}, d_{3,\omega}, id_{3,\omega}, soc_{3,\omega}$	$W_3, PV_3, \lambda_{3,\omega}^{IB\pm} \in \mathbb{R}$	$e_{3,\omega}^{IB+}, e_{3,\omega}^{IB-}$
8	$fd_4, afd_4^\pm, c_{4,\omega}, d_{4,\omega}, id_{4,\omega}, soc_{4,\omega}$	$W_4, PV_4, \lambda_{4,\omega}^{IB\pm} \in \mathbb{R}$	$e_{4,\omega}^{IB+}, e_{4,\omega}^{IB-}$
9	$fd_5, afd_5^\pm, c_{5,\omega}, d_{5,\omega}, id_{5,\omega}, soc_{5,\omega}$	$W_5, PV_5, \lambda_{5,\omega}^{IB\pm} \in \mathbb{R}$	$e_{5,\omega}^{IB+}, e_{5,\omega}^{IB-}$
10	$fd_6, afd_6^\pm, c_{6,\omega}, d_{6,\omega}, id_{6,\omega}, soc_{6,\omega}$	$W_6, PV_6, \lambda_{6,\omega}^{IB\pm} \in \mathbb{R}$	$e_{6,\omega}^{IB+}, e_{6,\omega}^{IB-}$
11	$fd_7, afd_7^\pm, c_{7,\omega}, d_{7,\omega}, id_{7,\omega}, soc_{7,\omega}$	$W_7, PV_7, \lambda_{7,\omega}^{IB\pm} \in \mathbb{R}$	$e_{7,\omega}^{IB+}, e_{7,\omega}^{IB-}$
12	$fd_8, afd_8^\pm, c_{8,\omega}, d_{8,\omega}, id_{8,\omega}, soc_{8,\omega}$	$W_8, PV_8, \lambda_{8,\omega}^{IB\pm} \in \mathbb{R}$	$p_{8,\omega}^{IB+}, p_{8,\omega}^{IB-}$
13	$fd_9, afd_9^\pm, c_{9,\omega}, d_{9,\omega}, id_{9,\omega}, soc_{9,\omega}$	$W_9, PV_9, \lambda_{9,\omega}^{IB\pm} \in \mathbb{R}$	$e_{9,\omega}^{IB+}, e_{9,\omega}^{IB-}$
14	$fd_{10}, afd_{10}^\pm, c_{10,\omega}, d_{10,\omega}, id_{10,\omega}, soc_{10,\omega}$	$W_{10}, PV_{10}, \lambda_{10,\omega}^{IB\pm} \in \mathbb{R}$	$e_{10,\omega}^{IB+}, e_{10,\omega}^{IB-}$
15	$e_{3,t,\omega}^{IM}$	$\lambda_3^I \in \mathbb{R}^{T_3}$	
16	$fd_{11}, afd_{11}^\pm, c_{11,\omega}, d_{11,\omega}, id_{11,\omega}, soc_{11,\omega}$	$W_{11}, PV_{11}, \lambda_{11,\omega}^{IB\pm} \in \mathbb{R}$	$e_{11,\omega}^{IB+}, e_{11,\omega}^{IB-}$
17	$fd_{12}, afd_{12}^\pm, c_{12,\omega}, d_{12,\omega}, id_{12,\omega}, soc_{12,\omega}$	$W_{12}, PV_{12}, \lambda_{12,\omega}^{IB\pm} \in \mathbb{R}$	$p_{12,\omega}^{IB+}, p_{12,\omega}^{IB-}$
18	$fd_{13}, afd_{13}^\pm, c_{13,\omega}, d_{13,\omega}, id_{13,\omega}, soc_{13,\omega}$	$W_{13}, PV_{13}, \lambda_{13,\omega}^{IB\pm} \in \mathbb{R}$	$p_{13,\omega}^{IB+}, p_{13,\omega}^{IB-}$
19	$fd_{14}, afd_{14}^\pm, c_{14,\omega}, d_{14,\omega}, id_{14,\omega}, soc_{14,\omega}$	$W_{14}, PV_{14}, \lambda_{14,\omega}^{IB\pm} \in \mathbb{R}$	$e_{14,\omega}^{IB+}, e_{14,\omega}^{IB-}$
20	$fd_{15}, afd_{15}^\pm, c_{15,\omega}, d_{15,\omega}, id_{15,\omega}, soc_{15,\omega}$	$W_{15}, PV_{15}, \lambda_{15,\omega}^{IB\pm} \in \mathbb{R}$	$e_{15,\omega}^{IB+}, e_{15,\omega}^{IB-}$
21	$fd_{16}, afd_{16}^\pm, c_{16,\omega}, d_{16,\omega}, id_{16,\omega}, soc_{16,\omega}$	$W_{16}, PV_{16}, \lambda_{16,\omega}^{IB\pm} \in \mathbb{R}$	$e_{16,\omega}^{IB+}, e_{16,\omega}^{IB-}$
22	$fd_{17}, afd_{17}^\pm, c_{17,\omega}, d_{17,\omega}, id_{17,\omega}, soc_{17,\omega}$	$W_{17}, PV_{17}, \lambda_{17,\omega}^{IB\pm} \in \mathbb{R}$	$e_{17,\omega}^{IB+}, e_{17,\omega}^{IB-}$
23	$fd_{18}, afd_{18}^\pm, c_{18,\omega}, d_{18,\omega}, id_{18,\omega}, soc_{18,\omega}$	$W_{18}, PV_{18}, \lambda_{18,\omega}^{IB\pm} \in \mathbb{R}$	$e_{18,\omega}^{IB+}, e_{18,\omega}^{IB-}$
24	$fd_{19}, afd_{19}^\pm, c_{19,\omega}, d_{19,\omega}, id_{19,\omega}, soc_{19,\omega}$	$W_{19}, PV_{19}, \lambda_{19,\omega}^{IB\pm} \in \mathbb{R}$	$e_{19,\omega}^{IB+}, e_{19,\omega}^{IB-}$
25	$fd_{20}, afd_{20}^\pm, c_{20,\omega}, d_{20,\omega}, id_{20,\omega}, soc_{20,\omega}$	$W_{20}, PV_{20}, \lambda_{20,\omega}^{IB\pm} \in \mathbb{R}$	$e_{20,\omega}^{IB+}, e_{20,\omega}^{IB-}$
26	$fd_{21}, afd_{21}^\pm, c_{21,\omega}, d_{21,\omega}, id_{21,\omega}, soc_{21,\omega}$	$W_{21}, PV_{21}, \lambda_{21,\omega}^{IB\pm} \in \mathbb{R}$	$e_{21,\omega}^{IB+}, e_{21,\omega}^{IB-}$
27	$fd_{22}, afd_{22}^\pm, c_{22,\omega}, d_{22,\omega}, id_{22,\omega}, soc_{22,\omega}$	$W_{22}, PV_{22}, \lambda_{22,\omega}^{IB\pm} \in \mathbb{R}$	$e_{22,\omega}^{IB+}, e_{22,\omega}^{IB-}$
28	$fd_{23}, afd_{23}^\pm, c_{23,\omega}, d_{23,\omega}, id_{23,\omega}, soc_{23,\omega}$	$W_{23}, PV_{23}, \lambda_{23,\omega}^{IB\pm} \in \mathbb{R}$	$e_{23,\omega}^{IB+}, e_{23,\omega}^{IB-}$
29	$fd_{24}, afd_{24}^\pm, c_{24,\omega}, d_{24,\omega}, id_{24,\omega}, soc_{24,\omega}$	$W_{24}, PV_{24}, \lambda_{24,\omega}^{IB\pm} \in \mathbb{R}$	$e_{24,\omega}^{IB+}, e_{24,\omega}^{IB-}$

Table 1.1: Summary of stages, decision variables, random variable observations and recourse variables.

1.2 15 minute resolution model maintaining number of stages

In this adaptation of the original model, we use a 15-minute resolution for all variables and parameters while keeping the same number of stages. That is, market bids are submitted for each subperiod of 15 minutes, but decisions are made only at the beginning of each hour.

Information is only revealed at the hourly stage, yet we retain detailed data for each 15-minute subperiod.

Therefore, the DAM, RM, IM IB and operational variables are indexed in T (period) and Q (subperiod).

1.2.1 Fundamental Elements

Sets:

$\mathcal{T}$ (T)	set of T time periods $\{1, \dots, T\}$ [hours/15-min periods].
$\mathcal{T}_0$ (T_0)	extended set of time periods $\{0, \dots, T\}$ [hours/15-min periods].
$\mathcal{Q}$ (Q)	set of Q subperiods per time period $\{1, \dots, Q\}$. IM, IB and operational variables are indexed in T and Q .
$\mathcal{I}$ (IM)	set of I intraday markets $\{1, \dots, I\}$.
$\mathcal{T}_i$ (TIM)	set of time periods in intraday market $i \in \mathcal{I}$ [hours/15-min periods].
$\mathcal{I}_t$ (IMT)	set of intraday markets where $t \in \mathcal{T}$ is a time period.
Ω (S)	set of scenarios.
$\mathcal{SG}$ (SG)	set of stages $\{1, \dots, nSG\}$.
$\mathcal{SG}_0$ (SG_0)	extended set of stages $\{0, \dots, nSG\}$.
$\mathcal{C}_{sg,\omega}$ (c)	scenario cluster of scenario $\omega \in \Omega$ at stage $sg \in \mathcal{SG}_0$.
$\mathcal{RV}$ (RV)	set of random variables $\{1, \dots, nRV\}$.

Parameters:

T (nT)	number of time periods [24 hours/96 15-min periods].
ΔT (dT)	duration of the period [1 / 0.25 hours].
Q (nQ)	number of time sub-periods (4 for quarterly hour resolution).
ΔQ (dQ)	duration of the subperiod [hours].
I (nIM)	number of intraday markets.
nSG (nSG)	number of stages (= T).
SG_i^{IM} ($sgim$)	stage in which the price of intraday market $i \in \mathcal{I}$ is revealed.
nRV (nRV)	number of random variables.
$nRVSG_{sg}$ ($nRVSG$)	number of random variables revealed at stage $sg \in \mathcal{SG}$.
Π_ω ($Prob$)	probability of scenario $\omega \in \Omega$.
$S_{rv,\omega}$ ($Scen$)	value of random variable $rv \in \mathcal{RV}$ in scenario $\omega \in \Omega$.

1.2.2 Flexible Demand

Sets:

$\mathcal{F}$ (FI) set of time intervals where a fraction of demand c_f has to be met.

Parameters:

nF (nFI) number of intervals in $\mathcal{F}$.
 $D_{t,q}$ (FD) central demand at $t \in \mathcal{T}$, $q \in \mathcal{Q}$ [MWh].
 $\underline{D}_{t,q}$ (FD_L) minimum demand at $t \in \mathcal{T}$, $q \in \mathcal{Q}$ (inflexible demand) [MWh].
 $\overline{D}_{t,q}$ (FD_U) maximum demand at $t \in \mathcal{T}$, $q \in \mathcal{Q}$ [MWh].
 T_f^- (TF_L) first time step of interval $f \in \mathcal{F}$ [hour].
 T_f^+ (TF_U) last time step of interval $f \in \mathcal{F}$ [hour].
 c_f (coef_FD) fraction of demand that needs to be consumed during interval $f \in \mathcal{F}$.
 C^{FD} (C_FD) cost per unit of flexible demand used [€/MWh].

Variables:

$fd_{t,q,\omega}$ (var_fd) flexible demand at $t \in \mathcal{T}$, $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ (after flexibility is considered) [MWh].
 $afd_{t,q,\omega}^+$ (var_afd_p) auxiliary variable: positive part of flexible demand shift from central demand $D_{t,q}$ [MWh].
 $afd_{t,q,\omega}^-$ (var_afd_m) auxiliary variable: negative part of flexible demand shift from central demand $D_{t,q}$ [MWh].

Nonnegativity constraints to the flexible demand variables:

$$fd_{t,q,\omega} \geq 0, afd_{t,q,\omega}^+ \geq 0, afd_{t,q,\omega}^- \geq 0 \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega. \quad (1.58)$$

Flexible demand bounds:

$$fd_{t,q,\omega} \leq \overline{D}_{t,q} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega \quad (1.59)$$

$$\underline{D}_{t,q} \leq fd_{t,q,\omega} \quad \forall t \in \mathcal{T}, q \in \mathcal{Q}, \forall \omega \in \Omega. \quad (1.60)$$

Total daily demand (DailyDem):

$$\sum_{t \in \mathcal{T}} \sum_{q \in \mathcal{Q}} fd_{t,q,\omega} = \sum_{t \in \mathcal{T}} \sum_{q \in \mathcal{Q}} D_{t,q} \quad \forall \omega \in \Omega. \quad (1.61)$$

Certain percentage of the central demand needs to be met over a time interval (InterDem):

$$\sum_{T_f^- \leq t \leq T_f^+} \sum_{q \in \mathcal{Q}} fd_{t,q,\omega} \geq c_f \sum_{T_f^- \leq t \leq T_f^+} \sum_{q \in \mathcal{Q}} D_{t,q} \quad \forall f \in \mathcal{F}, \forall \omega \in \Omega. \quad (1.62)$$

Split the variables $fd_{t,w}$ into its positive and negative parts (FlexDemDisplace):

$$D_{t,q} - fd_{t,q,\omega} = afd_{t,q,\omega}^+ - afd_{t,q,\omega}^- \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega. \quad (1.63)$$

1.2.3 Wind Farm

Parameters:

SG_t^{WP} (sgpw)	stage in which the wind production at time $t \in \mathcal{T}$ is revealed.
$W_{t,q,\omega}$ (pW)	wind production at time $t \in \mathcal{T}$, subperiod $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [MWh].
$\overline{P}^W$ (max_pW)	wind nameplate capacity [MW].

1.2.4 Solar Farm

Parameters:

$PV_{t,q,\omega}$ (pPV)	solar production at time $t \in \mathcal{T}$, subperiod $q \in \mathcal{Q}$, under scenario $\omega \in \Omega$ [MWh].
$\overline{P}^{PV}$ (max_pPV)	solar nameplate capacity [MW].

1.2.5 Battery Energy Storage System

Sets:

Parameters:

E^B (Emax)	BESS energy capacity [MWh].
P^B (Dmax)	BESS maximum charging/discharging rate [MW].
η^B (RTE)	BESS round-trip efficiency.
$\overline{\text{SoC}}$ (SOCmax)	BESS maximum state of charge. Unitless, values in $[0, 1]$.
$\underline{\text{SoC}}$ (SOCmin)	BESS minimum state of charge. Unitless, values in $[0, 1]$.
SoC_0 (SOCini)	BESS initial state of charge. Unitless, values in $[0, 1]$.
SoC_T (SOCfin)	BESS final state of charge. Unitless, values in $[0, 1]$.

Variables:

$c_{t,q,\omega}$ (cV)	BESS charge rate at time $t \in \mathcal{T}$, $q \in \mathcal{Q}$, under scenario $\omega \in \Omega$ [MW].
$d_{t,q,\omega}$ (dV)	BESS discharge rate at time $t \in \mathcal{T}$, $q \in \mathcal{Q}$, under scenario $\omega \in \Omega$ [MW].
$id_{t,q,\omega}$ (idV)	binary. Indicates the charging ($id_{t,q,\omega} = 0$) or discharging ($id_{t,q,\omega} = 1$) state of the BESS at time $t \in \mathcal{T}$, $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$.
$\text{soc}_{t,q,\omega}$ (socV)	state of charge of BESS at time $t \in \mathcal{T}_0$, $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$. Unitless, values in $[0, 1]$.

Nonnegativity constraints:

$$c_{t,q,\omega} \geq 0, \quad d_{t,q,\omega} \geq 0, \quad id_{t,q,\omega} \in \{0, 1\} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q} \forall \omega \in \Omega. \quad (1.64)$$

Discharging/charging bounds (VPP_state_dV and VPP_state_cV):

$$d_{t,q,\omega} \leq P^B id_{t,q,\omega} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega \quad (1.65)$$

$$c_{t,q,\omega} \leq P^B (1 - id_{t,q,\omega}) \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega. \quad (1.66)$$

Variable state of charge $\text{soc}_{t,q,\omega}$ of the BESS (SOCV):

$$\text{soc}_{t,q,\omega} = \begin{cases} \text{soc}_{t-1,Q,\omega} + \frac{\Delta Q \left(c_{t,q,\omega} - \frac{d_{t,q,\omega}}{\eta^B} \right)}{E^B}, & q = 1, \\ \text{soc}_{t,q-1,\omega} + \frac{\Delta Q \left(c_{t,q,\omega} - \frac{d_{t,q,\omega}}{\eta^B} \right)}{E^B}, & q = 2, \dots, Q, \end{cases} \quad \forall t \in \mathcal{T}, \forall \omega \in \Omega. \quad (1.67)$$

Operational limits:

$$\underline{\text{SoC}} \leq \text{soc}_{t,q,\omega} \leq \overline{\text{SoC}} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega. \quad (1.68)$$

Initial and final states of charge of the BESS (SOCV_ini and SOCV_fin):

$$\text{soc}_{0,Q,\omega} = \text{SoC}_0 \quad \forall \omega \in \Omega \quad (1.69)$$

$$\text{soc}_{T,Q,\omega} = \text{SoC}_T \quad \forall \omega \in \Omega. \quad (1.70)$$

1.2.6 Market Participation

Day Ahead Market

Sets:

$\mathcal{SSD}$ (Ssd) $\{(t, q, l, j) \in \mathcal{T} \times \mathcal{Q} \times \Omega \times \Omega \mid l \neq j, \lambda_{t,q,l}^D \leq \lambda_{t,q,j}^D\}$. Rather technical, For every $t \in \mathcal{T}$ and $q \in \mathcal{Q}$ provides the scenario pairs (l, q, j) s.t. $\lambda_{t,q,l}^D \leq \lambda_{t,q,j}^D$ (order is important). Used to ensure monotonicity of day-ahead bids w.r.t. day-ahead price realisations.

Parameters:

$\lambda_{t,q,\omega}^D$ (1D) day-ahead market price at time $t \in \mathcal{T}$ $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [€/MWh].

$\underline{E}^{DA}$ (PDA_LB) minimum bid size to day-ahead market [MWh].

Variables:

$e_{t,q,\omega}^{DA+}$ (eDA_p) energy matched in the day-ahead market of a selling bid by EC at time $t \in \mathcal{T}$ $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [MWh].

$e_{t,q,\omega}^{DA-}$ (eDA_m) energy matched in the day-ahead market of a buying bid by EC at time $t \in \mathcal{T}$ $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [MWh].

$ie_{t,q,\omega}^{DA+}$ (ieDA_p) binary. $ie_{t,q,\omega}^{DA+} = 1$ if the EC sells electricity at time $t \in \mathcal{T}$ and $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$, $ie_{t,q,\omega}^{DA+} = 0$ otherwise.

$ie_{t,q,\omega}^{DA-}$ (ieDA_m) binary. $ie_{t,q,\omega}^{DA-} = 1$ if the EC buys electricity at time $t \in \mathcal{T}$ and $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$, $ie_{t,q,\omega}^{DA-} = 0$ otherwise.

Nonnegativity constraints:

$$e_{t,q,\omega}^{DA+} \geq 0, e_{t,q,\omega}^{DA-} \geq 0, ie_{t,q}^{DA+}, ie_{t,q}^{DA-} \in \{0, 1\} \quad \forall t \in \mathcal{T} \quad \forall q \in \mathcal{Q} \quad \forall \omega \in \Omega. \quad (1.71)$$

Bounds on the selling (DA_sell_bid_LB and DA_sell_bid_UB) and buying (DA_buy_bid_LB and DA_buy_bid_UB) energy matched quantities:

$$\underline{E}^{DA} ie_{t,q,\omega}^{DA+} \leq e_{t,q,\omega}^{DA+} \leq (\bar{P}^W \Delta Q + \bar{P}^{PV} \Delta Q + P^B \Delta Q - \underline{D}_{t,q}) ie_{t,q,\omega}^{DA+} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \quad \forall \omega \in \Omega. \quad (1.72)$$

$$\underline{E}^{DA} ie_{t,q,\omega}^{DA-} \leq e_{t,q,\omega}^{DA-} \leq (P^B \Delta Q + \bar{D}_{t,q}) ie_{t,q,\omega}^{DA-} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \quad \forall \omega \in \Omega. \quad (1.73)$$

$$ie_{t,q,\omega}^{DA+} + ie_{t,q,\omega}^{DA-} \leq 1 \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \quad \forall \omega \in \Omega. \quad (1.74)$$

Offer curves need to be monotone w.r.t. the scenario prices (DA_bid_mono_1 and DA_bid_mono_2)

$$e_{t,q,\omega}^{DA+} \leq e_{t,q,\omega'}^{DA+} \quad \forall (t, q, \omega, \omega') \in \mathcal{SSD} \quad (1.75)$$

$$e_{t,q,\omega}^{DA-} \geq e_{t,q,\omega'}^{DA-} \quad \forall (t, q, \omega, \omega') \in \mathcal{SSD}. \quad (1.76)$$

Reserve Market

Parameters:

$\lambda_{t,q,\omega}^R$ (1R)	reserve market price at time $t \in \mathcal{T}$ $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [€/MWh].
T^R (TSR)	duration of reserve response (fixed) [h].
$R_{t,q}^{U,FD}$ (RUFD)	upper bound on the upwards reserve provided by flexible demand [MW].
$R_{t,q}^{D,FD}$ (RDFD)	upper bound on the downwards reserve provided by flexible demand [MW].

Variables:

$r_{t,q,\omega}^U$ (rU)	upward reserve matched by the EC at time $t \in \mathcal{T}$ $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [MW].
$r_{t,q,\omega}^D$ (rD)	downward reserve matched by the EC at time $t \in \mathcal{T}$ $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [MW].
$r_{t,q,\omega}^{U,B}$ (rU_B)	BESS component of upward reserve matched at time $t \in \mathcal{T}$ $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [MW].
$r_{t,q,\omega}^{D,B}$ (rD_B)	BESS component of downward reserve matched at time $t \in \mathcal{T}$ $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [MW].
$r_{t,q,\omega}^{U,FD}$ (rU_FD)	flexible demand component of upward reserve matched at time $t \in \mathcal{T}$ $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [MW].
$r_{t,q,\omega}^{D,FD}$ (rD_FD)	flexible demand component of downward reserve matched at time $t \in \mathcal{T}$ $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [MW].

Nonnegativity constraints:

$$r_{t,q,\omega}^U \geq 0, \quad r_{t,q,\omega}^D \geq 0, \quad r_{t,q,\omega}^{U,B} \geq 0, \quad r_{t,q,\omega}^{D,B} \geq 0, \quad r_{t,q,\omega}^{U,FD} \geq 0, \quad r_{t,q,\omega}^{D,FD} \geq 0, \quad \forall u \in \mathcal{U}, t \in \mathcal{T}, q \in \mathcal{Q} \text{ and } \omega \in \Omega. \quad (1.77)$$

Components of the RM (RM_components_U and RM_components_D):

$$r_{t,q,\omega}^U = r_{t,q,\omega}^{U,B} + r_{t,q,\omega}^{U,FD} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q} \quad \forall \omega \in \Omega \quad (1.78)$$

$$r_{t,q,\omega}^D = r_{t,q,\omega}^{D,B} + r_{t,q,\omega}^{D,FD} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q} \quad \forall \omega \in \Omega. \quad (1.79)$$

Enough energy for the demand given the RM (`FlexDem_UBand FlexDem_LB`):

$$fd_{t,q,\omega} + T^R r_{t,q,\omega}^{D,FD} \leq \overline{D}_{t,q} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q} \quad \forall \omega \in \Omega \quad (1.80)$$

$$\underline{D}_{t,q} \leq fd_{t,q,\omega} - T^R r_{t,q,\omega}^{U,FD} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega. \quad (1.81)$$

Bounds for the RM coming from flexible demand (`FlexDemP_DReserve` and `FlexDemP_UReserve`):

$$r_{t,q,\omega}^{D,FD} \leq R_{t,q}^{D,FD} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega \quad (1.82)$$

$$r_{t,q,\omega}^{U,FD} \leq R_{t,q}^{U,FD} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega. \quad (1.83)$$

Bounds for the RM coming from the BESS (`VPP_RU2s` and `VPP_RD2s`):

$$r_{t,q,\omega}^{U,B} - c_{t,q,\omega} + d_{t,q,\omega} \leq P^B \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega \quad (1.84)$$

$$r_{t,q,\omega}^{D,B} + c_{t,q,\omega} - d_{t,q,\omega} \leq P^B \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega. \quad (1.85)$$

Bounds on the state of the battery including reserve (`VPP_RU_SOCV` and `VPP_RD_SOCV`):

$$\underline{\text{SoC}} \leq \text{soc}_{t,q,\omega} - \frac{T^R r_{t,q,\omega}^{U,B}}{\eta^B E^B} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega \quad (1.86)$$

$$\text{soc}_{t,q,\omega} + \frac{T^R r_{t,q,\omega}^{D,B}}{E^B} \leq \overline{\text{SoC}} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega. \quad (1.87)$$

Intraday Markets

Parameters:

$\lambda_{i,t,q,\omega}^I$ (`1I`) intraday market $i \in \mathcal{I}$ price at time $t \in \mathcal{T}$, subperiod $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [€/MWh].

R^{IDA} (`maxTIM`) bound on the ratio of energy traded in intraday markets and day ahead market (to avoid speculation).

Variables:

$e_{i,t,q,\omega}^{IM}$ (`eIM`) energy matched by the EC at intraday market $i \in \mathcal{I}$ at time $t \in \mathcal{T}$, $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [MWh].

Constraint to control speculative behaviour (`IM_bounds_1`, `IM_bounds_2`, `IM_bounds_3` and `IM_bounds_4`):

$$-R^{IDA}(e_{t,q,\omega}^{DA+} + e_{t,q,\omega}^{DA-}) \leq \sum_{i \in \mathcal{I}_t} e_{i,t,q,\omega}^{IM} \leq R^{IDA}(e_{t,q,\omega}^{DA+} + e_{t,q,\omega}^{DA-}) \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega \quad (1.88)$$

$$-R^{IDA}(e_{t,q,\omega}^{DA+} + e_{t,q,\omega}^{DA-}) \leq e_{i,t,q,\omega}^{IM} \leq R^{IDA}(e_{t,q,\omega}^{DA+} + e_{t,q,\omega}^{DA-}) \quad \forall i \in \mathcal{I}, \forall t \in \mathcal{T}_i, \forall q \in \mathcal{Q}, \forall \omega \in \Omega. \quad (1.89)$$

System Imbalances

Parameters:

- $\lambda_{t,q,\omega}^{IB+}$ (1PIB) positive imbalance price at time $t \in \mathcal{T}$ $q \in \mathcal{Q}$ under scenario $\omega \in \Omega$ [€/MWh].
- $\lambda_{t,q,\omega}^{IB-}$ (1NIB) negative imbalance price at time $t \in \mathcal{T}$, $q \in \mathcal{Q}$, under scenario $\omega \in \Omega$ [€/MWh].
- $E_{t,q,\omega}^{IB+}$ (PIB_p) upper bound on the positive imbalance at time $t \in \mathcal{T}$, $q \in \mathcal{Q}$, under scenario $\omega \in \Omega$ [MWh].
- $E_{t,q,\omega}^{IB-}$ (PIB_m) upper bound on the negative imbalance at time $t \in \mathcal{T}$, $q \in \mathcal{Q}$, under scenario $\omega \in \Omega$ [MWh].

Variables:

- $e_{t,q,\omega}^{IB-}$ (pIB_m) negative imbalance of EC at time $t \in \mathcal{T}$, $q \in \mathcal{Q}$, under scenario $\omega \in \Omega$ [MWh].
- $e_{t,q,\omega}^{IB+}$ (pIB_p) positive imbalance of EC at time $t \in \mathcal{T}$, $q \in \mathcal{Q}$, under scenario $\omega \in \Omega$ [MWh].

Nonnegativity constraints:

$$e_{t,q,\omega}^{IB+} \geq 0, \quad e_{t,q,\omega}^{IB-} \geq 0 \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q} \forall \omega \in \Omega. \quad (1.90)$$

The imbalances of the EC are defined in the following equation (**Imbalances**)

$$e_{t,q,\omega}^{IB+} - e_{t,q,\omega}^{IB-} = e_{t,q,\omega}^{DA-} + W_{t,q,\omega} + PV_{t,q,\omega} + d_{t,q,\omega} \Delta Q - \left(e_{t,q,\omega}^{DA+} + \sum_{i \in \mathcal{I}_t} e_{i,t,q,\omega}^{IM} + f d_{t,q,\omega} + c_{t,q,\omega} \Delta Q \right) \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega \quad (1.91)$$

Control for speculative behaviour (IB_pos_UB and IB_neg_UB):

$$e_{t,q,\omega}^{IB+} \leq E_{t,q,\omega}^{IB+} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega \quad (1.92)$$

$$e_{t,q,\omega}^{IB-} \leq E_{t,q,\omega}^{IB-} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega. \quad (1.93)$$

1.2.7 Nonanticipativity Constraints

For DA matched energy variables (NAC_eDA_p and NAC_eDA_m)

$$e_{t,q,\omega}^{DA+} = e_{t,q,\omega'}^{DA+} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega', \quad (1.94)$$

$$e_{t,q,\omega}^{DA-} = e_{t,q,\omega'}^{DA-} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega'. \quad (1.95)$$

For DA binary variables $ie_{t,\omega}^{DA+}$, we have (NAC_ieDA_p and NAC_ieDA_m):

$$ie_{t,q,\omega}^{DA+} = ie_{t,q,\omega'}^{DA+} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega'. \quad (1.96)$$

$$ie_{t,q,\omega}^{DA-} = ie_{t,q,\omega'}^{DA-} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega'. \quad (1.97)$$

For upward reserve variables, we have (NAC_rU and NAC_rU_B and NAC_rU_FD):

$$r_{t,q,\omega}^U = r_{t,q,\omega'}^U \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega', \quad (1.98)$$

$$r_{t,q,\omega}^{U,B} = r_{t,q,\omega'}^{U,B} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega', \quad (1.99)$$

$$r_{t,q,\omega}^{U,FD} = r_{t,q,\omega'}^{U,FD} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega'. \quad (1.100)$$

For downward reserve variables, we have (NAC_rD, NAC_rD_B and NAC_rD_FD))

$$r_{t,q,\omega}^D = r_{t,q,\omega'}^D \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega', \quad (1.101)$$

$$r_{t,q,\omega}^{D,B} = r_{t,q,\omega'}^{D,B} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega', \quad (1.102)$$

$$r_{t,q,\omega}^{D,FD} = r_{t,q,\omega'}^{D,FD} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{1,\omega}, \omega \neq \omega'. \quad (1.103)$$

For intraday market variables, we have (NAC_eIM)

$$e_{i,t,q,\omega}^{IM} = e_{i,t,q,\omega'}^{IM} \quad \forall i \in \mathcal{I}, \forall t \in \mathcal{T}_i, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_i^{IM}-1,\omega}, \omega \neq \omega'. \quad (1.104)$$

For imbalance variables, we have (NAC_pIB_p and NAC_pIB_m):

$$e_{t,q,\omega}^{IB+} = e_{t,q,\omega'}^{IB+} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP},\omega}, \omega \neq \omega', \quad (1.105)$$

$$e_{t,q,\omega}^{IB-} = e_{t,q,\omega'}^{IB-} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP},\omega}, \omega \neq \omega'. \quad (1.106)$$

For flexible demand variables, we have (NAC_var_fd, NAC_var_afd_p and NAC_var_afd_m):

$$fd_{t,q,\omega} = fd_{t,q,\omega'} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega', \quad (1.107)$$

$$afd_{t,q,\omega}^+ = afd_{t,q,\omega'}^+ \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega', \quad (1.108)$$

$$afd_{t,q,\omega}^- = afd_{t,q,\omega'}^- \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega'. \quad (1.109)$$

For BESS operational variables, we have (NAC_cV, NAC_dV, NAC_idV and NAC_socV):

$$c_{t,q,\omega} = c_{t,q,\omega'} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega', \quad (1.110)$$

$$d_{t,q,\omega} = d_{t,q,\omega'} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega', \quad (1.111)$$

$$id_{t,q,\omega} = id_{t,q,\omega'} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega', \quad (1.112)$$

$$soc_{t,q,\omega} = soc_{t,q,\omega'} \quad \forall t \in \mathcal{T}, \forall q \in \mathcal{Q}, \forall \omega \in \Omega, \forall \omega' \in \mathcal{C}_{SG_t^{WP}-1,\omega}, \omega \neq \omega'. \quad (1.113)$$

1.2.8 Objective Function

We define the objective function Expected Energy Community Social Welfare (EECSW) as follows:

$$EECSW := \sum_{t \in \mathcal{T}} \sum_{q \in \mathcal{Q}} \sum_{\omega \in \Omega} \Pi_{\omega} \lambda_{t,q,\omega}^D (e_{t,q,\omega}^{DA+} - e_{t,q,\omega}^{DA-}) \quad (1.114a)$$

$$+ \sum_{t \in \mathcal{T}} \sum_{q \in \mathcal{Q}} \sum_{\omega \in \Omega} \Pi_{\omega} \lambda_{t,q,\omega}^R (r_{t,q,\omega}^D + r_{t,q,\omega}^U) T^R \quad (1.114b)$$

$$+ \sum_{t \in \mathcal{T}} \sum_{q \in \mathcal{Q}} \sum_{\omega \in \Omega} \Pi_{\omega} \sum_{i \in \mathcal{I}_t} \lambda_{i,t,\omega}^I e_{i,t,\omega}^{IM} \quad (1.114c)$$

$$+ \sum_{t \in \mathcal{T}} \sum_{q \in \mathcal{Q}} \sum_{\omega \in \Omega} \Pi_{\omega} \lambda_{t,\omega}^{IB+} e_{t,\omega}^{IB+} \quad (1.114d)$$

$$- \sum_{t \in \mathcal{T}} \sum_{q \in \mathcal{Q}} \sum_{\omega \in \Omega} \Pi_{\omega} \lambda_{t,\omega}^{IB-} e_{t,\omega}^{IB-} \quad (1.114e)$$

$$- \sum_{t \in \mathcal{T}} \sum_{q \in \mathcal{Q}} \sum_{\omega \in \Omega} \Pi_{\omega} C^{FD} (af d_{t,\omega}^+ + af d_{t,\omega}^-). \quad (1.114f)$$

1.2.9 Complete Model Formulation

max	$EECSW$	
s.t.	(1.58), (1.61), (1.62), (1.63)	Flex. Demand Op.
	(1.64), (1.65), (1.66), (1.67), (1.68), (1.69), (1.70)	BESS Op.
	(1.71), (1.72), (1.73), (1.74), (1.75), (1.76)	EC DA
	(1.77), (1.78), (1.79), (1.80), (1.81), (1.82), (1.83), (1.84), (1.85), (1.86), (1.87)	EC RM
	(1.88), (1.89)	EC IM
	(1.90), (1.91), (1.92), (1.93)	EC IB
	(1.94), (1.95), (1.96), (1.97)	NAC DA
	(1.98), (1.99), (1.100), (1.101), (1.102), (1.103)	NAC RM
	(1.104)	NAC IM
	(1.105), (1.106)	NAC Imbalances
	(1.107), (1.108), (1.109)	NAC Flex. Demand
	(1.110), (1.111), (1.112), (1.113)	NAC BESS Op.

1.2.10 Summary of Stages for $T = 24$ (hourly) with subperiods

In Table 1.2, we present a summary of the stages considered. Each stage can contain decision variables, observations and recourse variables. Note that, in terms of the optimization model, recourse variables of one stage are equivalent to decision variables of the next stage.

Stage	Decision Variables	Observations	Recourse Variables
1		$\lambda^D \in \mathbb{R}^{T \times \mathcal{Q}}$	$e_{t,q,\omega}^{DA+}, e_{t,q,\omega}^{DA-}, ie_{t,q,\omega}^{DA+}, ie_{t,q,\omega}^{DA-}$
2	$r_{t,q,\omega}^U, r_{t,q,\omega}^{U,B}, r_{t,q,\omega}^{U,q,FD}, r_{t,q,\omega}^D, r_{t,q,\omega}^{D,B}, r_{t,q,\omega}^{D,FD}$	$\lambda^R \in \mathbb{R}^{T \times \mathcal{Q}}$	
3	$e_{1,t,q,\omega}^{IM}$	$\lambda_1^I \in \mathbb{R}^{T_1 \times \mathcal{Q}}$	
4	$e_{2,t,q,\omega}^{IM}$	$\lambda_2^I \in \mathbb{R}^{T_2 \times \mathcal{Q}}$	
5	$fd_{1,q}, afd_{1,q}^{\pm}, c_{1,q,\omega}, d_{1,q,\omega}, id_{1,q,\omega}, soc_{1,q,\omega}$	$W_1, PV_1, \lambda_{1,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{1,q,\omega}^{IB+}, e_{1,q,\omega}^{IB-}$
6	$fd_{2,q}, afd_{2,q}^{\pm}, c_{2,q,\omega}, d_{2,q,\omega}, id_{2,q,\omega}, soc_{2,q,\omega}$	$W_2, PV_2, \lambda_{2,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{2,q,\omega}^{IB+}, e_{2,q,\omega}^{IB-}$
7	$fd_{3,q}, afd_{3,q}^{\pm}, c_{3,q,\omega}, d_{3,q,\omega}, id_{3,q,\omega}, soc_{3,q,\omega}$	$W_3, PV_3, \lambda_{3,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{3,q,\omega}^{IB+}, e_{3,q,\omega}^{IB-}$
8	$fd_{4,q}, afd_{4,q}^{\pm}, c_{4,q,\omega}, d_{4,q,\omega}, id_{4,q,\omega}, soc_{4,q,\omega}$	$W_4, PV_4, \lambda_{4,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{4,q,\omega}^{IB+}, e_{4,q,\omega}^{IB-}$
9	$fd_{5,q}, afd_{5,q}^{\pm}, c_{5,q,\omega}, d_{5,q,\omega}, id_{5,q,\omega}, soc_{5,q,\omega}$	$W_5, PV_5, \lambda_{5,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{5,q,\omega}^{IB+}, e_{5,q,\omega}^{IB-}$
10	$fd_{6,q}, afd_{6,q}^{\pm}, c_{6,q,\omega}, d_{6,q,\omega}, id_{6,q,\omega}, soc_{6,q,\omega}$	$W_6, PV_6, \lambda_{6,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{6,q,\omega}^{IB+}, e_{6,q,\omega}^{IB-}$
11	$fd_{7,q}, afd_{7,q}^{\pm}, c_{7,q,\omega}, d_{7,q,\omega}, id_{7,q,\omega}, soc_{7,q,\omega}$	$W_7, PV_7, \lambda_{7,\omega}^{IB+} \in \mathbb{R}^{\mathcal{Q}}$	$e_{7,q,\omega}^{IB+}, e_{7,q,\omega}^{IB-}$
12	$fd_{8,q}, afd_{8,q}^{\pm}, c_{8,q,\omega}, d_{8,q,\omega}, id_{8,q,\omega}, soc_{8,q,\omega}$	$W_8, PV_8, \lambda_{8,\omega}^{IB+} \in \mathbb{R}^{\mathcal{Q}}$	$p_{8,q,\omega}^{IB+}, p_{8,q,\omega}^{IB-}$
13	$fd_{9,q}, afd_{9,q}^{\pm}, c_{9,q,\omega}, d_{9,q,\omega}, id_{9,q,\omega}, soc_{9,q,\omega}$	$W_9, PV_9, \lambda_{9,\omega}^{IB+} \in \mathbb{R}^{\mathcal{Q}}$	$e_{9,q,\omega}^{IB+}, e_{9,q,\omega}^{IB-}$
14	$fd_{10,q}, afd_{10,q}^{\pm}, c_{10,q,\omega}, d_{10,q,\omega}, id_{10,q,\omega}, soc_{10,q,\omega}$	$W_{10}, PV_{10}, \lambda_{10,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{10,q,\omega}^{IB+}, e_{10,q,\omega}^{IB-}$
15	$e_{3,t,q,\omega}^{IM}$	$\lambda_3^I \in \mathbb{R}^{T_3 \times \mathcal{Q}}$	
16	$fd_{11,q}, afd_{11,q}^{\pm}, c_{11,q,\omega}, d_{11,q,\omega}, id_{11,q,\omega}, soc_{11,q,\omega}$	$W_{11}, PV_{11}, \lambda_{11,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{11,q,\omega}^{IB+}, e_{11,q,\omega}^{IB-}$
17	$fd_{12,q}, afd_{12,q}^{\pm}, c_{12,q,\omega}, d_{12,q,\omega}, id_{12,q,\omega}, soc_{12,q,\omega}$	$W_{12}, PV_{12}, \lambda_{12,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$p_{12,q,\omega}^{IB+}, p_{12,q,\omega}^{IB-}$
18	$fd_{13,q}, afd_{13,q}^{\pm}, c_{13,q,\omega}, d_{13,q,\omega}, id_{13,q,\omega}, soc_{13,q,\omega}$	$W_{13}, PV_{13}, \lambda_{13,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$p_{13,q,\omega}^{IB+}, p_{13,q,\omega}^{IB-}$
19	$fd_{14,q}, afd_{14,q}^{\pm}, c_{14,q,\omega}, d_{14,q,\omega}, id_{14,q,\omega}, soc_{14,q,\omega}$	$W_{14}, PV_{14}, \lambda_{14,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{14,q,\omega}^{IB+}, e_{14,q,\omega}^{IB-}$
20	$fd_{15,q}, afd_{15,q}^{\pm}, c_{15,q,\omega}, d_{15,q,\omega}, id_{15,q,\omega}, soc_{15,q,\omega}$	$W_{15}, PV_{15}, \lambda_{15,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{15,q,\omega}^{IB+}, e_{15,q,\omega}^{IB-}$
21	$fd_{16,q}, afd_{16,q}^{\pm}, c_{16,q,\omega}, d_{16,q,\omega}, id_{16,q,\omega}, soc_{16,q,\omega}$	$W_{16}, PV_{16}, \lambda_{16,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{16,q,\omega}^{IB+}, e_{16,q,\omega}^{IB-}$
22	$fd_{17,q}, afd_{17,q}^{\pm}, c_{17,q,\omega}, d_{17,q,\omega}, id_{17,q,\omega}, soc_{17,q,\omega}$	$W_{17}, PV_{17}, \lambda_{17,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{17,q,\omega}^{IB+}, e_{17,q,\omega}^{IB-}$
23	$fd_{18,q}, afd_{18,q}^{\pm}, c_{18,q,\omega}, d_{18,q,\omega}, id_{18,q,\omega}, soc_{18,q,\omega}$	$W_{18}, PV_{18}, \lambda_{18,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{18,q,\omega}^{IB+}, e_{18,q,\omega}^{IB-}$
24	$fd_{19,q}, afd_{19,q}^{\pm}, c_{19,q,\omega}, d_{19,q,\omega}, id_{19,q,\omega}, soc_{19,q,\omega}$	$W_{19}, PV_{19}, \lambda_{19,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{19,q,\omega}^{IB+}, e_{19,q,\omega}^{IB-}$
25	$fd_{20,q}, afd_{20,q}^{\pm}, c_{20,q,\omega}, d_{20,q,\omega}, id_{20,q,\omega}, soc_{20,q,\omega}$	$W_{20}, PV_{20}, \lambda_{20,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{20,q,\omega}^{IB+}, e_{20,q,\omega}^{IB-}$
26	$fd_{21,q}, afd_{21,q}^{\pm}, c_{21,q,\omega}, d_{21,q,\omega}, id_{21,q,\omega}, soc_{21,q,\omega}$	$W_{21}, PV_{21}, \lambda_{21,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{21,q,\omega}^{IB+}, e_{21,q,\omega}^{IB-}$
27	$fd_{22,q}, afd_{22,q}^{\pm}, c_{22,q,\omega}, d_{22,q,\omega}, id_{22,q,\omega}, soc_{22,q,\omega}$	$W_{22}, PV_{22}, \lambda_{22,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{22,q,\omega}^{IB+}, e_{22,q,\omega}^{IB-}$
28	$fd_{23,q}, afd_{23,q}^{\pm}, c_{23,q,\omega}, d_{23,q,\omega}, id_{23,q,\omega}, soc_{23,q,\omega}$	$W_{23}, PV_{23}, \lambda_{23,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{23,q,\omega}^{IB+}, e_{23,q,\omega}^{IB-}$
29	$fd_{24,q}, afd_{24,q}^{\pm}, c_{24,q,\omega}, d_{24,q,\omega}, id_{24,q,\omega}, soc_{24,q,\omega}$	$W_{24}, PV_{24}, \lambda_{24,\omega}^{IB\pm} \in \mathbb{R}^{\mathcal{Q}}$	$e_{24,q,\omega}^{IB+}, e_{24,q,\omega}^{IB-}$

Table 1.2: Summary of stages, decision variables, random variable observations and recourse variables.

Bibliography

- [1] F.-Javier Heredia, Marlyn D. Cuadrado, and Cristina Corchero. “On optimal participation in the electricity markets of wind power plants with battery energy storage systems”. In: *Computers & Operations Research* 96 (2018), pages 316–329. DOI: 10.1016/j.cor.2018.03.004.